Nader Khalil

Full-Stack Software Engineer

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Professional Summary

Full-stack software engineer who builds web applications and AI assistants that people can rely on: easy to use, hard to break, and backed by automated tests. Software Engineering graduate, now training at 42 Abu Dhabi in C, C++, and Unix systems. Picks up new tools quickly and works well both individually and in a team.

Technical Skills

Frontend: React, Next.js, TypeScript, Vite, TanStack Query, Tailwind CSS, accessible UI (axe)

Backend: REST APIs, layered architecture, Zod, Auth.js, PHP, C# MinimalAPI

Data & Security: PostgreSQL + pgvector, MariaDB, Prisma, row locks, CSP, CSRF, rate limiting

AI: LLM integration, tool calling with human review, RAG, embeddings, LLM evals, prompt caching, Vercel AI SDK

Testing & CI: Vitest, Playwright, PHPUnit, race tests, GitHub Actions, dependency audits, Docker

Languages: TypeScript, JavaScript, Python, C, C++, C#, Linux

Professional Experience

AI Trainer & Mentor, *Department of Education and Knowledge (ADEK)* 07/2026 – 08/2026

- Delivered technical instruction on agentic workflows, prompt structuring, and autonomous tool calling for the accredited AgenticVerse program.
- Mentored cohort teams building functional AI prototypes addressing real problem statements from ADNOC, FANR, and Khalifa Fund.
- Provided technical troubleshooting and support for participants experiencing environment setup issues and configuration errors.
- Guided students through code debugging, API schema validation, and structured output parsing, explaining technical concepts clearly so they could solve problems on their own.

QA & Software Tester, *42 Abu Dhabi* 01/2026

- Conducted functional and UI testing on an internal student platform using black-box testing and use-case scenarios to identify bugs before release.
- Tested features across browsers and screen sizes to catch responsive and layout issues before deployment.
- Re-tested fixes and checked related flows to confirm bugs were resolved without regressions.
- Wrote detailed bug reports with clear reproduction steps to help technical teams isolate and resolve issues quickly.

IT Support & Technical Staff, *Ticketmaster* 12/2025

- Provisioned and deployed specialized entry ticketing systems and scanner hardware for a high-volume international event.
- Served as the on-site technical point of contact, delivering rapid incident resolution under strict, time-sensitive conditions.
- Performed system health checks and connectivity audits, maintaining 100% operational uptime for entry points throughout the event.

Software Engineering Intern, *Timex IT Consulting* 01/2024 – 04/2024

- Built C# MinimalAPI endpoints supporting sorting, pagination, and file handling.
- Implemented request validation and standardized HTTP response handling across CRUD services.
- Supported software release rollouts by identifying and documenting functional bugs in proprietary accounting software.
- Coordinated with developers to verify technical fixes based on defect reports before client deployment.

Technical Projects

AI Booking Platform

Live demo: ai-booking-platform.onrender.com · Case study: naderkhalil.dev/case-study.html

- One booking engine for any appointment business, grown out of my AI Appointment Scheduler: clients book from a live calendar, advisors publish availability, and admins manage users, booking rules, and the knowledge base without a deploy. Next.js 16, TypeScript, PostgreSQL + pgvector, Prisma, Auth.js.
- AI assistant answers from the admin-managed knowledge base with citations, filtered by role in SQL, and prepares bookings and changes as cards that run only after the user confirms; about 120 live checks (prompt injection, data leaks, scope, Arabic) gate prompt changes.
- A business is configuration, not code: two sample businesses run on the same codebase, and a test fails if a business's words leak into the code.
- Concurrency-safe bookings (conditional updates and row locks, race-tested against a real Postgres), nonce-based CSP, Zod validation, rate limits, and 340 unit, 302 integration, and 84 end-to-end tests in CI.

Student-Faculty Services v2, *Al Ain University capstone rewrite*

- Rebuilt my 2024 senior capstone as a React 19, TypeScript, and PHP 8.4 app on MariaDB with 46 REST endpoints and no framework, deployed by GitHub Actions only when every test passes. Includes reminders, ratings, a waitlist, and a 10-second undo for cancellations.
- Prevented double bookings with ordered database row locks, proven by a race test where 8 students book one slot over real HTTP; 500+ PHPUnit, Vitest, and Playwright tests with axe accessibility checks (WCAG 2.1 AA).

42 Intra Logtime Calculator, *published Chrome extension*

- A lightweight extension with 60+ users that adds live logtime totals to the 42 Intra interface; it runs entirely client-side, collects no data, and is 19 KB.

Minishell (Unix Shell Implementation), *42 Abu Dhabi*

- Developed a functional command interpreter in C supporting pipelines, environment variable expansion, and stream redirection.
- Managed process lifecycles via fork(), execve(), and signal handling routines.
- Prevented memory leaks across dynamic allocations using strict Valgrind analysis.

Education & Certifications

B.Sc. in Software Engineering, *Al Ain University* 09/2019 – 05/2024

Computer Science Diploma, *42 Abu Dhabi* 12/2024 – Present

CyberQ CTF, Top 10 Finalist, *UAE Cybersecurity Council* 2025

Hack The Box University CTF, *119th of 1,014 teams* 2025

MOEI × 42 Abu Dhabi AgentEra AI Hackathon, *Certificate of Appreciation* 2026

AI on Watch with Datamind, *RecognAIze Hackathon* 2025

Languages

Arabic	**English**
Native	Professional Proficiency